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Batch:- 25

Lab Assignment (1-3)

**Multiplication Table Generator**

Write a program to input a number n and print its multiplication table up to 10 using all four loops (for, while, do-while, for-each).

Input:-

```java
import java.util.Scanner;

public class MultiplicationTable {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        // Using for loop
        System.out.println("\nMultiplication Table using FOR loop:");
        for (int i = 1; i <= 10; i++) {
            System.out.println(n + " x " + i + " = " + (n * i));
        }

        // Using while loop
        System.out.println("\nMultiplication Table using WHILE loop:");
        int i = 1;
        while (i <= 10) {
            System.out.println(n + " x " + i + " = " + (n * i));
            i++;
        }

        //  Using do-while loop
        System.out.println("\nMultiplication Table using DO-WHILE loop:");
        int j = 1;
        do {
            System.out.println(n + " x " + j + " = " + (n * j));
            j++;
        } while (j <= 10);

        // Using for-each loop
        System.out.println("\nMultiplication Table using FOR-EACH loop:");
        int[] arr = {1,2,3,4,5,6,7,8,9,10};
        for (int num : arr) {
            System.out.println(n + " x " + num + " = " + (n * num));
        }

        sc.close();
    }
}
```

Output :-

```
niyatijha@Niyatis-MacBook-Air java codes  % javac MultiplicationTable.java
niyatijha@Niyatis-MacBook-Air java codes  % java MultiplicationTable
Enter a number: 12

Multiplication Table using FOR loop:
12 x 1 = 12
12 x 2 = 24
12 x 3 = 36
12 x 4 = 48
12 x 5 = 60
12 x 6 = 72
12 x 7 = 84
12 x 8 = 96
12 x 9 = 108
12 x 10 = 120

Multiplication Table using WHILE loop:
12 x 1 = 12
12 x 2 = 24
12 x 3 = 36
12 x 4 = 48
12 x 5 = 60
12 x 6 = 72
12 x 7 = 84
12 x 8 = 96
12 x 9 = 108
12 x 10 = 120

Multiplication Table using DO-WHILE loop:
12 x 1 = 12
12 x 2 = 24
12 x 3 = 36
12 x 4 = 48
12 x 5 = 60
12 x 6 = 72
12 x 7 = 84
12 x 8 = 96
12 x 9 = 108
12 x 10 = 120

Multiplication Table using FOR-EACH loop:
12 x 1 = 12
12 x 2 = 24
12 x 3 = 36
12 x 4 = 48
12 x 5 = 60
12 x 6 = 72
12 x 7 = 84
12 x 8 = 96
12 x 9 = 108
12 x 10 = 120
niyatijha@Niyatis-MacBook-Air java codes  %
```

**Factorial of a Number**

Write a program to compute the factorial of a given number using all four loops.

```java
import java.util.Scanner;

public class Factorial {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        //  Using for loop
        long factFor = 1;
        for (int i = 1; i <= n; i++) {
            factFor *= i;
        }
        System.out.println("\nFactorial using FOR loop: " + factFor);

        // Using while loop
        long factWhile = 1;
        int i = 1;
        while (i <= n) {
            factWhile *= i;
            i++;
        }
        System.out.println("Factorial using WHILE loop: " + factWhile);

        // Using do-while loop
        long factDoWhile = 1;
        int j = 1;
        do {
            factDoWhile *= j;
            j++;
        } while (j <= n);
        System.out.println("Factorial using DO-WHILE loop: " + factDoWhile);

        // Using for-each loop
        long factForEach = 1;
        int[] arr = new int[n];
        for (int k = 0; k < n; k++) {
            arr[k] = k + 1;
        }

        for (int num : arr) {
            factForEach *= num;
        }
        System.out.println("Factorial using FOR-EACH loop: " + factForEach);

        sc.close();
    }
}

```

Output:-

```
use --help for a list of possible options
[niyatijha@Niyatis-MacBook-Air java codes  % javac Factorial.java
[niyatijha@Niyatis-MacBook-Air java codes  % java Factorial
Enter a number: 5

Factorial using FOR loop: 120
Factorial using WHILE loop: 120
Factorial using DO-WHILE loop: 120
Factorial using FOR-EACH loop: 120
niyatijha@Niyatis-MacBook-Air java codes  %
```

**Reverse a Number**

Write a program to reverse the digits of a number using for, while, do-while, and for-each.

Input :-

```java
import java.util.Scanner;

public class ReverseNumberAllLoops {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        // Using for loop
        int temp1 = n;
        int revFor = 0;
        for (; temp1 != 0; temp1 /= 10) {
            int digit = temp1 % 10;
            revFor = revFor * 10 + digit;
        }
        System.out.println("\nReverse using FOR loop: " + revFor);

        // Using while loop
        int temp2 = n;
        int revWhile = 0;
        while (temp2 != 0) {
            int digit = temp2 % 10;
            revWhile = revWhile * 10 + digit;
            temp2 /= 10;
        }
        System.out.println("Reverse using WHILE loop: " + revWhile);

        // Using do-while loop
        int temp3 = n;
        int revDoWhile = 0;
        do {
            int digit = temp3 % 10;
            revDoWhile = revDoWhile * 10 + digit;
            temp3 /= 10;
        } while (temp3 != 0);
        System.out.println("Reverse using DO-WHILE loop: " + revDoWhile);

        // Using for-each loop
        String s = String.valueOf(n);
        int revForEach = 0;
        int[] digits = new int[s.length()];

        for (int i = 0; i < s.length(); i++) {
            digits[i] = s.charAt(i) - '0';
        }
```

```java
        int revForEach = 0;
        int[] digits = new int[s.length()];

        for (int i = 0; i < s.length(); i++) {
            digits[i] = s.charAt(i) - '0';
        }

        for (int digit : digits) {
            revForEach = revForEach * 10 + digit;
        }

        // Now reverse the constructed number properly
        int finalRevForEach = 0;
        int temp = revForEach;
        while (temp != 0) {
            finalRevForEach = finalRevForEach * 10 + (temp % 10);
            temp /= 10;
        }

        System.out.println("Reverse using FOR-EACH loop: " + finalRevForEach);

        sc.close();
    }
}
```

Output:-

```
niyatijha@Niyatis-MacBook-Air java codes  % javac Reverse.java
niyatijha@Niyatis-MacBook-Air java codes  % java Reverse
Enter a number: 456

Reverse using FOR loop: 654
Reverse using WHILE loop: 654
Reverse using DO-WHILE loop: 654
Reverse using FOR-EACH loop: 654
niyatijha@Niyatis-MacBook-Air java codes  %
```

**Fibonacci Series**

Write a program to generate the first n terms of the Fibonacci sequence using all four loop types.

Input :-

```java
import java.util.Scanner;

public class FibonacciAllLoops {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of terms: ");
        int n = sc.nextInt();

        if (n <= 0) {
            System.out.println("Please enter a positive number.");
            return;
        }

        // Using for loop
        System.out.println("\nFibonacci Series using FOR loop:");
        int a = 0, b = 1;
        for (int i = 1; i <= n; i++) {
            System.out.print(a + " ");
            int c = a + b;
            a = b;
            b = c;
        }

        //  Using while loop
        System.out.println("\n\nFibonacci Series using WHILE loop:");
        int x = 0, y = 1;
        int i = 1;
        while (i <= n) {
            System.out.print(x + " ");
            int z = x + y;
            x = y;
            y = z;
            i++;
        }

        // Using do-while loop
        System.out.println("\n\nFibonacci Series using DO-WHILE loop:");
        int p = 0, q = 1;
        int j = 1;
        do {
            System.out.print(p + " ");
            int r = p + q;
            p = q;
            q = r;
            j++;
```

```java
        int j = 1;
        do {
            System.out.print(p + " ");
            int r = p + q;
            p = q;
            q = r;
            j++;
        } while (j <= n);

        // Using for-each loop
        System.out.println("\n\nFibonacci Series using FOR-EACH loop:");
        int[] fib = new int[n];
        if (n >= 1) fib[0] = 0;
        if (n >= 2) fib[1] = 1;

        for (int k = 2; k < n; k++) {
            fib[k] = fib[k - 1] + fib[k - 2];
        }

        for (int num : fib) {
            System.out.print(num + " ");
        }

        sc.close();
    }
}
```

Output:-

```
niyatijha@Niyatis-MacBook-Air java codes  % javac Fibonacci.java
niyatijha@Niyatis-MacBook-Air java codes  % java Fibonacci
 Enter number of terms: 7

 Fibonacci Series using FOR loop:
 0 1 1 2 3 5 8

 Fibonacci Series using WHILE loop:
 0 1 1 2 3 5 8

 Fibonacci Series using DO-WHILE loop:
 0 1 1 2 3 5 8

 Fibonacci Series using FOR-EACH loop:
 0 1 1 2 3 5 8 %
 niyatijha@Niyatis-MacBook-Air java codes  %
```

**Sum of Digits**

Write a program to calculate the sum of digits of a number using all loop types.

Input :-

```java
import java.util.Scanner;

public class Sum1{

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        // Using for loop
        int tempFor = n;
        int sumFor = 0;
        for (; tempFor != 0; tempFor /= 10) {
            sumFor += tempFor % 10;
        }
        System.out.println("\nSum of digits using FOR loop: " + sumFor);

        // Using while loop
        int tempWhile = n;
        int sumWhile = 0;
        while (tempWhile != 0) {
            sumWhile += tempWhile % 10;
            tempWhile /= 10;
        }
        System.out.println("Sum of digits using WHILE loop: " + sumWhile);

        // Using do-while loop
        int tempDo = n;
        int sumDoWhile = 0;
        do {
            sumDoWhile += tempDo % 10;
            tempDo /= 10;
        } while (tempDo != 0);
        System.out.println("Sum of digits using DO-WHILE loop: " + sumDoWhile);

        // Using for-each loop
        String s = String.valueOf(n);
        int[] digits = new int[s.length()];
        for (int i = 0; i < s.length(); i++) {
            digits[i] = s.charAt(i) - '0';
        }

        int sumForEach = 0;
        for (int digit : digits) {
            sumForEach += digit;
        }
```

```
        int[] digits = new int[s.length()];
        for (int i = 0; i < s.length(); i++) {
            digits[i] = s.charAt(i) - '0';
        }

        int sumForEach = 0;
        for (int digit : digits) {
            sumForEach += digit;
        }
        System.out.println("Sum of digits using FOR-EACH loop: " + sumForEach);

        sc.close();
    }
}
```

Output :-

```
niyatijha@Niyatis-MacBook-Air java codes  % javac Sum1.java
niyatijha@Niyatis-MacBook-Air java codes  % java Sum1
Enter a number: 4567

Sum of digits using FOR loop: 22
Sum of digits using WHILE loop: 22
Sum of digits using DO-WHILE loop: 22
Sum of digits using FOR-EACH loop: 22
niyatijha@Niyatis-MacBook-Air java codes  %
```

**Find Maximum and Minimum in an Array**

Write a program to input an array and determine the maximum and minimum element using for, while, do-while, and for-each.

Input:-

```java
import java.util.Scanner;

public class MaxMinAllLoops {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the size of the array: ");
        int n = sc.nextInt();

        int[] arr = new int[n];
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        // Using for loop
        int maxFor = arr[0];
        int minFor = arr[0];
        for (int i = 1; i < n; i++) {
            if (arr[i] > maxFor) maxFor = arr[i];
            if (arr[i] < minFor) minFor = arr[i];
        }
        System.out.println("\nUsing FOR loop:");
        System.out.println("Maximum: " + maxFor);
        System.out.println("Minimum: " + minFor);

        // Using while loop
        int maxWhile = arr[0];
        int minWhile = arr[0];
        int i = 1;
        while (i < n) {
            if (arr[i] > maxWhile) maxWhile = arr[i];
            if (arr[i] < minWhile) minWhile = arr[i];
            i++;
        }
        System.out.println("\nUsing WHILE loop:");
        System.out.println("Maximum: " + maxWhile);
        System.out.println("Minimum: " + minWhile);

        // Using do-while loop
        int maxDo = arr[0];
        int minDo = arr[0];
        int j = 1;
        if (n > 1) { // ensure array has more than 1 element
            do {
                if (arr[j] > maxDo) maxDo = arr[j];
```

```java
        // Using do-while loop
        int maxDo = arr[0];
        int minDo = arr[0];
        int j = 1;
        if (n > 1) { // ensure array has more than 1 element
            do {
                if (arr[j] > maxDo) maxDo = arr[j];
                if (arr[j] < minDo) minDo = arr[j];
                j++;
            } while (j < n);
        }
        System.out.println("\nUsing DO-WHILE loop:");
        System.out.println("Maximum: " + maxDo);
        System.out.println("Minimum: " + minDo);

        // Using for-each loop
        int maxForEach = arr[0];
        int minForEach = arr[0];
        for (int num : arr) {
            if (num > maxForEach) maxForEach = num;
            if (num < minForEach) minForEach = num;
        }
        System.out.println("\nUsing FOR-EACH loop:");
        System.out.println("Maximum: " + maxForEach);
        System.out.println("Minimum: " + minForEach);

        sc.close();
    }
}
```

Output:-

```
[niyatijha@Niyatis-MacBook-Air java codes  % javac MaxMin.java
[niyatijha@Niyatis-MacBook-Air java codes  % java MaxMin
 Enter the size of the array: 5
 Enter 5 elements:
 12 7 25 3 18

 Using FOR loop:
 Maximum: 25
 Minimum: 3

 Using WHILE loop:
 Maximum: 25
 Minimum: 3

 Using DO-WHILE loop:
 Maximum: 25
 Minimum: 3

 Using FOR-EACH loop:
 Maximum: 25
 Minimum: 3
 niyatijha@Niyatis-MacBook-Air java codes  %
```

**Prime Number Check**

Write a program to check whether a given number is prime or not using all loop variations.

Input :-

```java
import java.util.Scanner;

public class PrimeAllLoops {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        if (n <= 1) {
            System.out.println("Number must be greater than 1 to check for prime.");
            return;
        }

        // Using for loop
        boolean isPrimeFor = true;
        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                isPrimeFor = false;
                break;
            }
        }
        System.out.println("Using FOR loop: " + n + (isPrimeFor ? " is Prime" : " is Not Prime"));

        // Using while loop
        boolean isPrimeWhile = true;
        int i = 2;
        while (i <= n / 2) {
            if (n % i == 0) {
                isPrimeWhile = false;
                break;
            }
            i++;
        }
        System.out.println("Using WHILE loop: " + n + (isPrimeWhile ? " is Prime" : " is Not Prime"));

        // Using do-while loop
        boolean isPrimeDo = true;
        int j = 2;
        if (n > 2) {
            do {
                if (n % j == 0) {
                    isPrimeDo = false;
                    break;
                }
                j++;
            } while (j <= n / 2);
        }
        System.out.println("Using DO-WHILE loop: " + n + (isPrimeDo ? " is Prime" : " is Not Prime"));

        // Using for-each loop
        int size = n / 2 - 1 + 1;
        if (n <= 3) size = 0;
        int[] arr = new int[size];
        for (int k = 0; k < size; k++) {
            arr[k] = k + 2;
        }

        boolean isPrimeForEach = true;
        for (int num : arr) {
            if (n % num == 0) {
                isPrimeForEach = false;
                break;
            }
        }
        System.out.println("Using FOR-EACH loop: " + n + (isPrimeForEach ? " is Prime" : " is Not Prime"));

        sc.close();
    }
}
```

Output:-

```
[niyatijha@Niyatis-MacBook-Air java codes  % javac Prime.java
[niyatijha@Niyatis-MacBook-Air java codes  % java Prime
 Enter a number: 5
 Using FOR loop: 5 is Prime
 Using WHILE loop: 5 is Prime
 Using DO-WHILE loop: 5 is Prime
 Using FOR-EACH loop: 5 is Prime
 niyatijha@Niyatis-MacBook-Air java codes  %
```

**Palindrome Checker**

Write a program to check whether a string or number is a palindrome using different loops.

Input:-

```java
import java.util.Scanner;

public class PalindromeAllLoops {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string or number: ");
        String input = sc.nextLine();

        // Using for loop
        boolean isPalindromeFor = true;
        int length = input.length();
        for (int i = 0; i < length / 2; i++) {
            if (input.charAt(i) != input.charAt(length - 1 - i)) {
                isPalindromeFor = false;
                break;
            }
        }
        System.out.println("\nUsing FOR loop: " + (isPalindromeFor ? "Palindrome" : "Not Palindrome"));

        // Using while loop
        boolean isPalindromeWhile = true;
        int start = 0;
        int end = length - 1;
        while (start < end) {
            if (input.charAt(start) != input.charAt(end)) {
                isPalindromeWhile = false;
                break;
            }
            start++;
            end--;
        }
        System.out.println("Using WHILE loop: " + (isPalindromeWhile ? "Palindrome" : "Not Palindrome"));

        // Using do-while loop
        boolean isPalindromeDo = true;
        start = 0;
        end = length - 1;
        if (length > 1) {
            do {
                if (input.charAt(start) != input.charAt(end)) {
                    isPalindromeDo = false;
                    break;
                }
                start++;
                end--;
```

```java
                break;
            }
            start++;
            end--;
        } while (start < end);
    }
    System.out.println("Using DO-WHILE loop: " + (isPalindromeDo ? "Palindrome" : "Not Palindrome"));

    // Using for-each loop
    char[] chars = input.toCharArray();
    int left = 0;
    int right = chars.length - 1;
    boolean isPalindromeForEach = true;
    for (char c : chars) {
        if (chars[left] != chars[right]) {
            isPalindromeForEach = false;
            break;
        }
        left++;
        right--;
        if (left >= right) break;
    }
    System.out.println("Using FOR-EACH loop: " + (isPalindromeForEach ? "Palindrome" : "Not Palindrome"));

    sc.close();
  }
}
```

Output:-

```
niyatijha@Niyatis-MacBook-Air java codes  % javac Palindrome.java
niyatijha@Niyatis-MacBook-Air java codes  % java Palindrome
Enter a string or number: 22

Using FOR loop: Palindrome
Using WHILE loop: Palindrome
Using DO-WHILE loop: Palindrome
Using FOR-EACH loop: Palindrome
niyatijha@Niyatis-MacBook-Air java codes  %
```

**Matrix Traversal**

Write a program to input a 2D matrix and print its elements row-wise and column-wise using for, while, do-while, and for-each.

Input :-

```java
import java.util.Scanner;

public class MatrixTraversal {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();
        System.out.print("Enter number of columns: ");
        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        System.out.println("Enter the elements of the matrix:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        // Row-wise traversal using for loop
        System.out.println("\nRow-wise (FOR loop):");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                System.out.print(matrix[i][j] + " ");
            }
            System.out.println();
        }

        // Row-wise traversal using while loop
        System.out.println("\nRow-wise (WHILE loop):");
        int i = 0;
        while (i < rows) {
            int j = 0;
            while (j < cols) {
                System.out.print(matrix[i][j] + " ");
                j++;
            }
            System.out.println();
            i++;
        }

        // Row-wise traversal using do-while loop
        System.out.println("\nRow-wise (DO-WHILE loop):");
        i = 0;
```

```java
            do {
                System.out.print(matrix[i][j] + " ");
                j++;
            } while (j < cols);
            System.out.println();
            i++;
        } while (i < rows);
    }

    // Row-wise traversal using for-each loop
    System.out.println("\nRow-wise (FOR-EACH loop):");
    for (int[] row : matrix) {
        for (int val : row) {
            System.out.print(val + " ");
        }
        System.out.println();
    }

    // Column-wise traversal using for loop
    System.out.println("\nColumn-wise (FOR loop):");
    for (int j = 0; j < cols; j++) {
        for (i = 0; i < rows; i++) {
            System.out.print(matrix[i][j] + " ");
        }
        System.out.println();
    }

    // Column-wise traversal using while loop
    System.out.println("\nColumn-wise (WHILE loop):");
    int j = 0;
    while (j < cols) {
        i = 0;
        while (i < rows) {
            System.out.print(matrix[i][j] + " ");
            i++;
        }
        System.out.println();
        j++;
    }

    // Column-wise traversal using do-while loop
    System.out.println("\nColumn-wise (DO-WHILE loop):");
    j = 0;
    if (rows > 0 && cols > 0) {
        do {
            i = 0;
            do {
                System.out.print(matrix[i][j] + " ");
```

```java
        if (rows > 0 && cols > 0) {
            do {
                i = 0;
                do {
                    System.out.print(matrix[i][j] + " ");
                    i++;
                } while (i < rows);
                System.out.println();
                j++;
            } while (j < cols);
        }

        // Column-wise traversal using for-each loop
        System.out.println("\nColumn-wise (FOR-EACH loop):");
        for (int col = 0; col < cols; col++) {
            for (int[] row : matrix) {
                System.out.print(row[col] + " ");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

Output:-

```
Using FOR-EACH loop: Palindrome
niyatijha@Niyatis-MacBook-Air java codes  % javac MatrixTraversal.java
niyatijha@Niyatis-MacBook-Air java codes  % java MatrixTraversal
Enter number of rows: 1 2 3
Enter number of columns: Enter the elements of the matrix:
4 5 6

Row-wise (FOR loop):
3 4

Row-wise (WHILE loop):
3 4

Row-wise (DO-WHILE loop):
3 4

Row-wise (FOR-EACH loop):
3 4

Column-wise (FOR loop):
3
4

Column-wise (WHILE loop):
3
4

Column-wise (DO-WHILE loop):
3
4

Column-wise (FOR-EACH loop):
3
4
niyatijha@Niyatis-MacBook-Air java codes  %
```

**Pattern Printing**

Write a program to print the following pattern using all loops:

Input:-

```java
import java.util.Scanner;

public class PatternPrintingAllLoops {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int n = sc.nextInt();

        // Using for loop
        System.out.println("\nPattern using FOR loop:");
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }

        // Using while loop
        System.out.println("\nPattern using WHILE loop:");
        int i = 1;
        while (i <= n) {
            int j = 1;
            while (j <= i) {
                System.out.print("*");
                j++;
            }
            System.out.println();
            i++;
        }

        // Using do-while loop
        System.out.println("\nPattern using DO-WHILE loop:");
        i = 1;
        if (n > 0) {
            do {
                int j = 1;
                do {
                    System.out.print("*");
                    j++;
                } while (j <= i);
                System.out.println();
                i++;
            } while (i <= n);
        }

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            } while (i <= n);
        }

        // Using for-each loop
        System.out.println("\nPattern using FOR-EACH loop:");
        int[] rowsArray = new int[n];
        for (int k = 0; k < n; k++) {
            rowsArray[k] = k + 1;
        }
        for (int row : rowsArray) {
            for (int j = 1; j <= row; j++) {
                System.out.print("*");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

Output:-

```
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niyatijha@Niyatis-MacBook-Air java codes  % javac Pattern.java
niyatijha@Niyatis-MacBook-Air java codes  % java Pattern
Enter number of rows: 5

Pattern using FOR loop:
*
**
***
****
*****

Pattern using WHILE loop:
*
**
***
****
*****

Pattern using DO-WHILE loop:
*
**
***
****
*****

Pattern using FOR-EACH loop:
*
**
***
****
*****
niyatijha@Niyatis-MacBook-Air java codes  %
```